

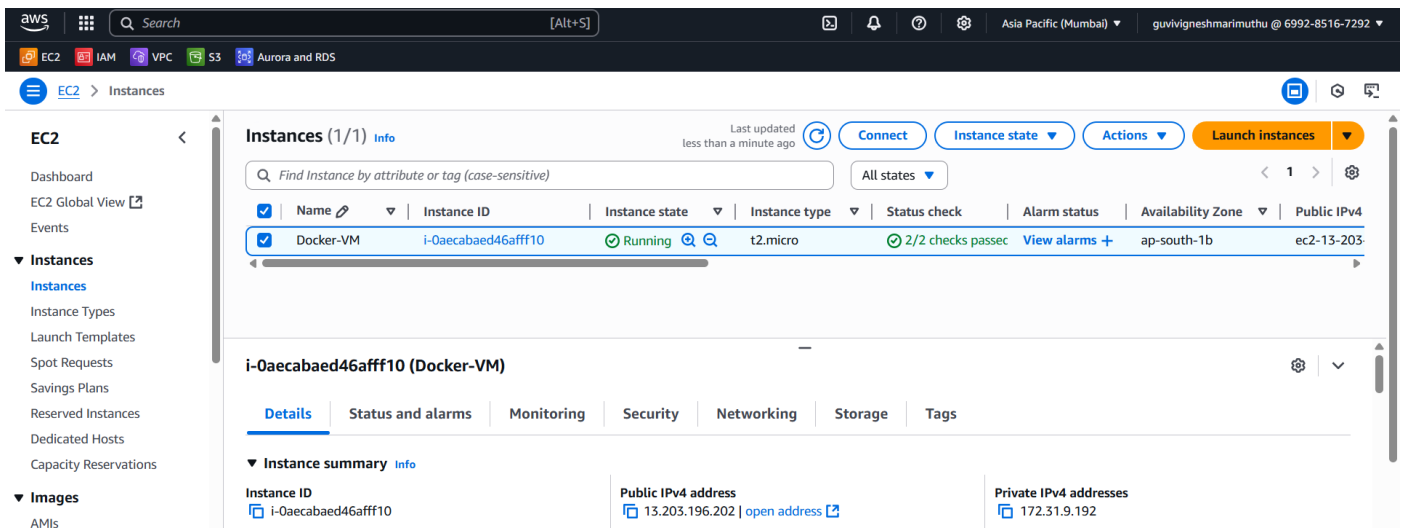
Docker Task

Task Description:

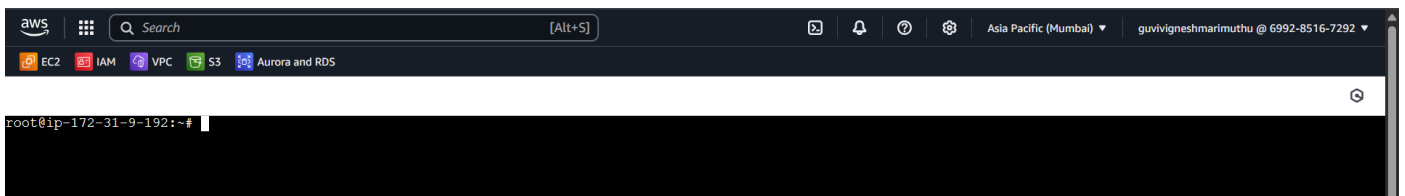
Install docker on EC2 and explore the docker commands (docker images, containers, volumes, network).

Solution:

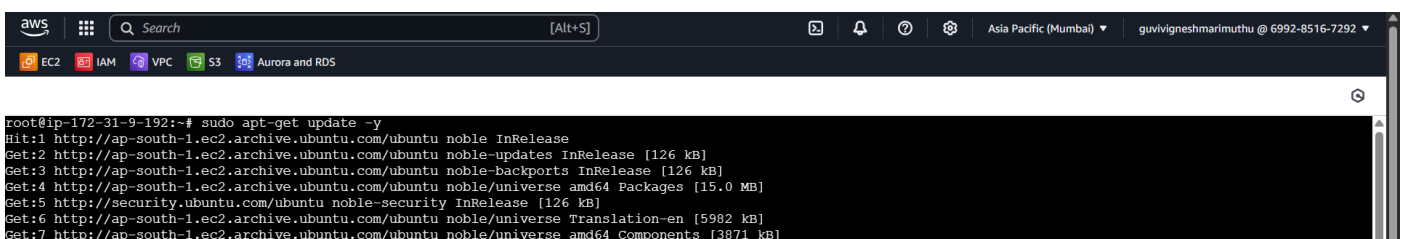
✓ 1. Launched a New EC2 Instance



✓ 2. Connected to the EC2 Instance Using SSH



✓ 3. Updated the Package Manager



✓ 4. Installed Docker



✓ 5. Started Docker Service & Checked Docker Version

```
root@ip-172-31-9-192:~# service docker start
root@ip-172-31-9-192:~# docker --version
Docker version 27.5.1, build 27.5.1-0ubuntu3~24.04.2
root@ip-172-31-9-192:~#
```

i-Oaecabaed46afff10 (Docker-VM)

PublicIPs: 13.203.196.202 PrivateIPs: 172.31.9.192

✓ 6. Explored Docker Commands

• Checked Docker Info

```
root@ip-172-31-9-192:~# docker info
Client:
Version:      27.5.1
Context:      default
Debug Mode:   false

Server:
Containers: 0
Running: 0
Paused: 0
Stopped: 0
Images: 0
Server Version: 27.5.1
```

• Pulled a Docker Image

```
root@ip-172-31-9-192:~# docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
dad67da3f26b: Pull complete
3b00567da964: Pull complete
56b81cfa547d: Pull complete
1bc5dc8b475d: Pull complete
979e6233a40a: Pull complete
d2a7ba8dbfee: Pull complete
32e44235e1d5: Pull complete
Digest: sha256:6784fb0834aa7dbbe12e3d7471e69c290df3e6ba810dc38b34ae33d3c1c05f7d
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
root@ip-172-31-9-192:~#
```

```
root@ip-172-31-9-192:~# docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
dad67da3f26b: Already exists
d0a755bf09a1: Pull complete
4f4fb700ef54: Pull complete
be5c5a616c3a: Pull complete
d1042d58e186: Pull complete
c06cec1379c2: Pull complete
Digest: sha256:f6557a77ee2f16c50a5ccbb2564a3fd56087da311bf69a160d43f73b23d3af2d
Status: Downloaded newer image for httpd:latest
docker.io/library/httpd:latest
root@ip-172-31-9-192:~#
```

• Listed All Docker Images

```
root@ip-172-31-9-192:~# docker images
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
nginx        latest   1e5f3c5b981a   2 months ago   192MB
httpd        latest   bf07fec943ec   5 months ago   148MB
root@ip-172-31-9-192:~#
```

i-Oaecabaed46afff10 (Docker-VM)

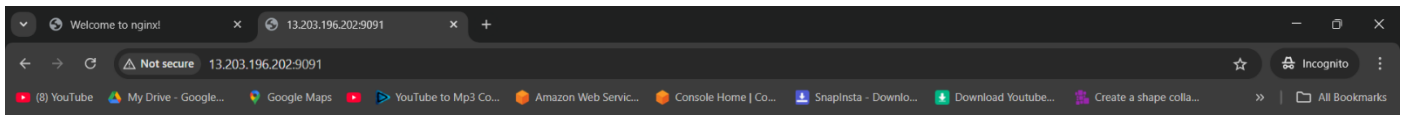
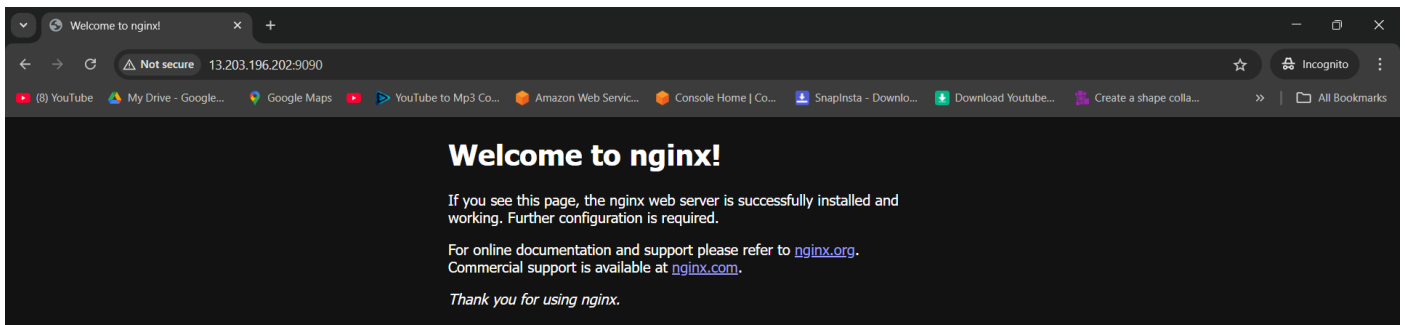
PublicIPs: 13.203.196.202 PrivateIPs: 172.31.9.192

• Created Containers & Checked Running Containers

```
root@ip-172-31-9-192:~# docker run -itd -p "9091:80" httpd
570eea613647b4efc1f461f2b693ff5c9ceba55058767e387ed0e1f677950b45
root@ip-172-31-9-192:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
570eea613647   httpd     "httpd-foreground"      7 seconds ago    Up 6 seconds    0.0.0.0:9091->80/tcp, [::]:9091->80/tcp    cranky_saha
11fddd4d87f0   nginx     "/docker-entrypoint..." 2 minutes ago    Up 2 minutes    0.0.0.0:9090->80/tcp, [::]:9090->80/tcp    nifty_ride
root@ip-172-31-9-192:~#
```

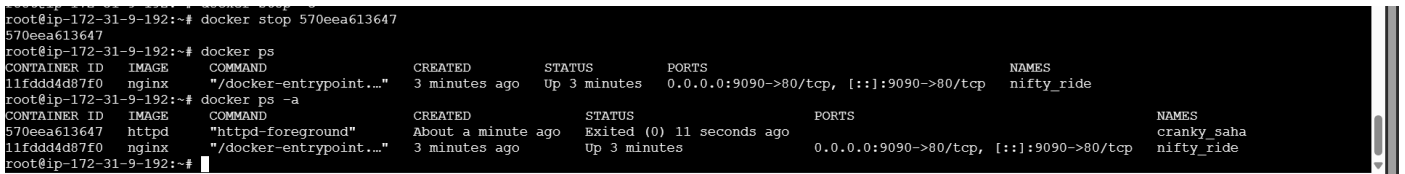
i-Oaecabaed46afff10 (Docker-VM)

PublicIPs: 13.203.196.202 PrivateIPs: 172.31.9.192

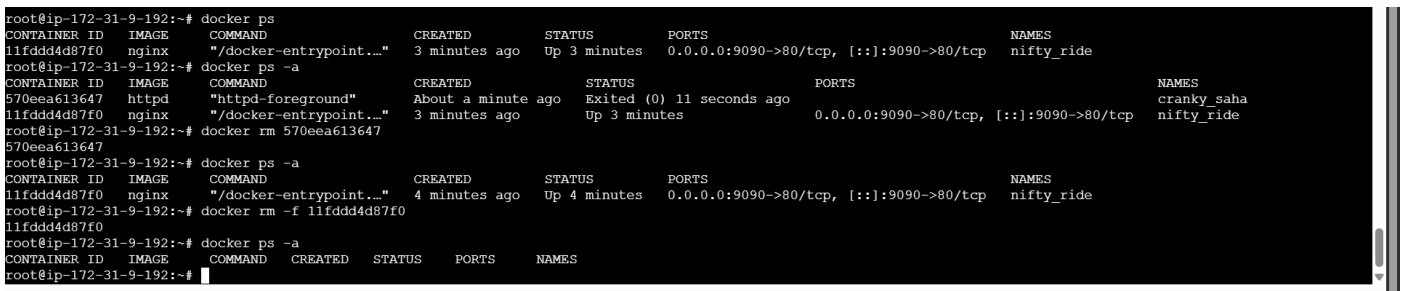


It works!

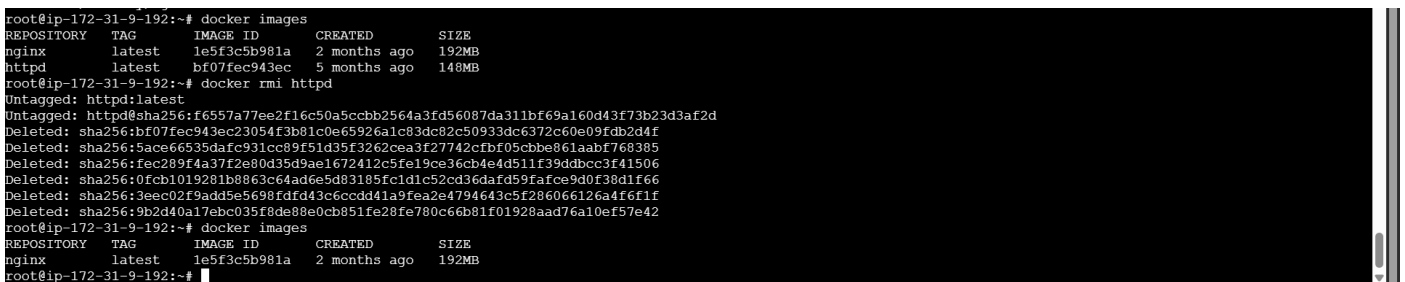
• Stopping Container & Checked All Containers (Including Stopped)



• Deleting Stopped Container & Running Container



• Deleting Images



• Explored Docker Networks



- **Explored Docker Volumes**

```
root@ip-172-31-9-192:~# docker volume ls
DRIVER      VOLUME NAME
root@ip-172-31-9-192:~# docker volume create myvolume
myvolume
root@ip-172-31-9-192:~# docker volume ls
DRIVER      VOLUME NAME
local       myvolume
root@ip-172-31-9-192:~#
```

i-Oaecabaed46afff10 (Docker-VM)

PublicIPs: 13.203.196.202 PrivateIPs: 172.31.9.192

- I explored and ran Docker commands for images, containers, volumes, and networks.
 - All commands were tested and verified with output.
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